

Delhi Public School, Sector-4, Bokaro

Subject: Physics

Class: VIII

Time Allowed: 45 minutes

Maximum Marks: 30

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class / Section: _____

SECTION A: MULTIPLE CHOICE QUESTIONS**Multiple Choice Questions**

Attempt all questions. Each question carries 1 mark.

1. Easy Which of the following is a unit of force? [1 Marks]
2. Easy The force of friction always acts in which direction to the motion of an object? [1 Marks]
3. Moderate Which of the following is an example of a non-contact force? [1 Marks]
4. Easy What is the SI unit of pressure? [1 Marks]
5. Easy Which of the following surfaces would offer the least friction? [1 Marks]
6. Easy The loudness of sound is determined by its: [1 Marks]
7. Moderate What type of lens is used to correct nearsightedness (myopia)? [1 Marks]
8. Moderate Which color of light bends the most when it passes through a prism? [1 Marks]
9. Moderate What is the process of transferring charge by touching a charged object to a neutral object called? [1 Marks]
10. Easy Which of the following is a renewable source of energy? [1 Marks]

SECTION B: SHORT ANSWER QUESTIONS

Short Answer Questions

Attempt all questions. Each question carries 2 marks.

11. Easy Define force and state its SI unit. [2 Marks]
12. Moderate Explain the difference between contact and non-contact forces. Give one example of each. [2 Marks]
13. Moderate What is friction? List two advantages and two disadvantages of friction. [2 Marks]
14. Moderate Explain the difference between loudness and pitch of sound. [2 Marks]
15. Moderate What is reflection of light? State the laws of reflection. [2 Marks]

SECTION C: DIAGRAM/GRAPH-BASED QUESTION

Diagram/Graph-Based

Attempt the following question. It carries 5 marks.

16. Challenging Draw a labeled diagram of the human eye. Explain the function of the lens and the retina. [5 Marks]

SECTION D: NUMERICAL PROBLEMS

Numerical Problems

Attempt the following question. It carries 5 marks.

17. Moderate A force of 20 N is applied to an object of mass 5 kg. Calculate the acceleration produced in the object. (Use the formula: Force = mass x acceleration) [5 Marks]

End of Question Paper

Answer Key

1. Newton
2. Opposite direction
3. Magnetic force
4. Pascal
5. Smooth ice
6. Amplitude
7. Concave lens
8. Blue
9. Conduction
10. Solar Energy
11. Force is a push or pull upon an object resulting from the object's interaction with another object. SI unit is Newton (N).
12. Contact forces require physical contact (e.g., friction). Non-contact forces don't (e.g., gravity).
13. Friction is a force that opposes motion. Advantages: walking, writing. Disadvantages: wear and tear, energy loss.
14. Loudness is the perception of sound intensity (amplitude). Pitch is the perception of sound frequency.
15. Reflection is the bouncing back of light. Laws: angle of incidence = angle of reflection; incident ray, reflected ray, and normal lie in the same plane.
16. Diagram should show cornea, iris, pupil, lens, retina, optic nerve. Lens focuses light on the retina. Retina contains light-sensitive cells that convert light into electrical signals.
17. Acceleration = Force / mass = $20 \text{ N} / 5 \text{ kg} = 4 \text{ m/s}^2$